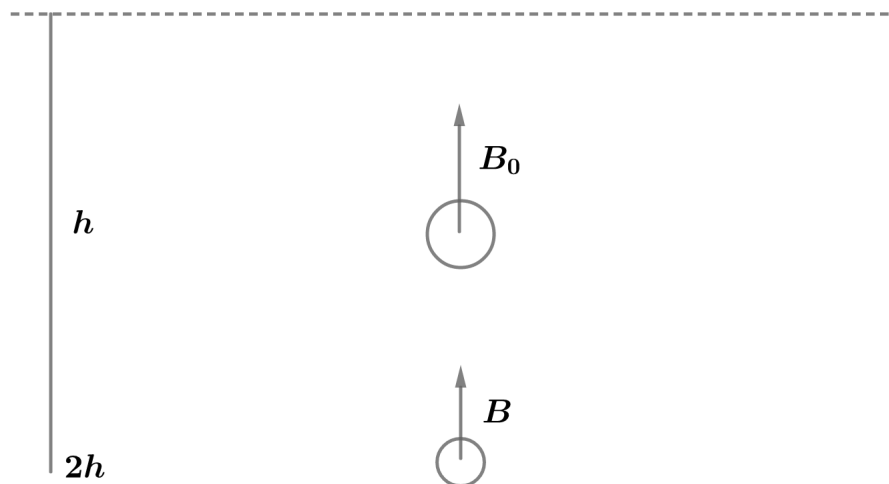


2018B F=ma Exam: Problem 10

Kevin S. Huang



Recall by Archimedes' principle that the buoyant force is given by

$$B = \rho_w V_{dis} g$$

Since water pressure increases as

$$P = \rho_w g h$$

and the balloon is compressible, we see that V_{dis} must decrease. Hence,

$$B < B_0$$

so the answer is D.